

Data Adaptive Filtering Approach to Improve the Classification Accuracy of Motor Imagery for BCI

Sanjoy Kumar Saha^{1,*} and Md. Sujan Ali^{2,3}

¹Department of Computer Science and Engineering, Begum Rokeya University
Rangpur, Bangladesh

²Signal Processing and Computational Neuroscience Lab, Department of Computer Science and Engineering
University of Rajshahi, Rajshahi, Bangladesh

³Department of Computer Science and Engineering, Jatiya Kabi Kazi Nazrul Islam University
Mymensingh, Bangladesh
*csesanjoy@gmail.com

Abstract—This paper presents a data adaptive pre-filtering method to enhance the motor imagery (MI) classification for brain computer interface (BCI) implementation electroencephalography (EEG). The filtering is implemented using multivariate empirical mode decomposition (MEMD). The MEMD decomposes multivariate EEG signal into a finite set of time varying basis functions called intrinsic mode functions (IMFs). Several IMFs are suppressed to improve the intelligibility of the EEG for better classification. The features from pre-filtered EEG are extracted by using common spatial pattern (CSP) which is a spatial filtering approach. The two class MI classification is performed by support vector machine (SVM). The proposed method is evaluated by BCI competition III of dataset IVa. Its performance is higher compared to that of the recently developed algorithm.

Index Terms—Brain computer interfaces (BCI), electroencephalogram (EEG), multivariate empirical mode decomposition (MEMD), common spatial pattern (CSP), support vector machine (SVM).

I. INTRODUCTION

A Brain-Computer Interfaces (BCI) which provides direct communication between human and computer based on human neurophysiological signals without depending on any muscle or peripheral nervous system activities [1]. Recently BCI based researches have been drawn attention in the field of brain science, neural engineering and rehabilitation [1]. In particular, motor imaginary (MI) based BCI is the imagination of a motor action without any real limbs movements, has practical significance in the case of lock in and spinal cord injured patients to communicate with their family, doctor or others. A class of electroencephalogram (EEG)-based BCIs which rely on MI of users are of particular interest to the BCI community since this type of BCI has a relatively robust communication performance, detectable on the scalp non-invasively with affordable equipment [2].

Different research shows that spatial distribution of EEG signals is different for different imagined movements [3]. For successful BCI implementation, it is necessary to identify the features of EEG signals and then convert them into control commands. It has been shown in [4] that motor imagery could cause the energy of alpha (7-12Hz) and beta (13-30Hz) rhythms increased or decreased, which has been known as event-related synchronization (ERS) or desynchronization (ERD). Nevertheless, the truth is the brain does not produce tones with fixed frequencies and additionally EEG come with noise due to several causes like bad electrode location,

electrode impedances, etc [5]. Although many methods have been reported for classification but a MI-based BCI system is not satisfactory due to the lack of classification accuracy. To tackle this problem Long Zhang et al. in [6] proposed different methods, where the EEG signal is decomposed into IMF using MEMD, and classified by CSP. Among different methods in [6] no single method showed better result over the entire subjects. The goal of the study is to improve the classification accuracy compared with reported methods in Long Zhang et al. in [6].

The proposed method focuses on the study of features extraction using data adaptive methods with common spatial pattern (CSP). EEG signals give an unclear image of brain activity because of the volume conduction. For this reason, CSP is used for discriminating between different classes of EEG activities based on their spatial configuration across electrodes. Common pattern (CSP) has been widely used in spatial filtering because of its efficiency and to improve BCI's performance [7] [8]. Empirical mode decomposition (EMD) is a fully data adaptive technique for analyzing nonlinear and non stationary signals [9]. The specialty of the method is that it decomposes the signal into intrinsic mode functions (IMFs) and the signal is reconstructed simply by summing up the bases with a negligible error. Recently it is reported in [10], its mode mixing is evident between intrinsic mode functions (IMFs) or a single mode is "leaked" into two IMFs. Bivariate and trivariate EMD (BEMD and TEMD) are also extension of EMD whereas performance depends on selecting variables in the case of multivariate data. By considering this problem, N. Rehman et al. [10] proposed multivariate EMD, is extension of standard EMD for analyzing multivariate data. The novelty of the method is that it reduces the mode mixing problem of the basis functions by taking signal projections along different directions in multi-dimensional spaces. The performance of CSP can be improved by decomposing EEG signal into IMFs and summing up optimal IMFs between 7 to 30 Hz due to smoothing CSP spatial filter with meaningful information. In our study, MEMD based common spatial pattern (MEMD-CSP) is presented to improve the performance of MI-BCI. The IMFs obtain using MEMD omitted based on examination. First IMF and last two IMFs are omitted because of, first IMF contains dominant frequency greater than 30 Hz and last two IMFs are residue.

After reconstructing signal, feature vector is obtained by CSP is classified using support vector machine (SVM). The

SVM utilizes a nonlinear map to translation the row training data into a higher dimension. Inside this new dimension, it seeks out for the linear optimal separating hyper-plane MI data from two classes can always be separated via a hyper-plane [11].

II. MULTIVARIATE EMPIRICAL MODE DECOMPOSITION (MEMD)

The Empirical Mode Decomposition (EMD) algorithm is a fully data-driven method designed for multi-scale decomposition and time-frequency analysis of real-world signals. Despite empirical mode decomposition (EMD) becoming a standard for time-frequency analysis of nonlinear and non-stationary signals, but it necessary to extend EMD for direct multichannel EEG data analysis. In EMD IMFs comes from different channel having a different number of IMFs, means they do not have the same frequency. To solve above problem multivariate EMD proposed by Rehman and Mandic [10] that a generic extension. Basic EMD find local mean by computing the average of upper and lower envelopes. Local mean of n -dimensional in MEMD method signals cannot be computed directly. Multiple n -dimensional envelopes are generated by projecting the signal along different directions in n -variant spaces [10]; those projections are then averaged to acquire the local mean. This step is complex to perform due to the lack of formal definition of maxima and minima in n -dimensional domains in general EMD. The sampling based on low discrepancy Hammersley sequence is used to generate projections of input signal in [10]. Once the projections along different directions in multidimensional spaces are obtained, their extrema are interpolated via cubic spline interpolation to obtain multiple-signal envelopes. Thus, obtained envelopes are then averaged to obtain the local mean of the multivariate signal. The following algorithm proposed in [10] is employed here to decompose signal $x(t)$ into a set of IMF components.

- I. Generate the point-set-based on the Hammersley sequence for sampling on an $(n-1)$ -sphere.
- II. Calculate a projection, denoted by $\{P^{\theta_k}(t)\}_{t=1}^T$ (here, $\{\theta_1, \theta_2, \dots, \theta_{n-1}\}$ be a set of the $(n-1)$ angular coordinates), of the input signal $\{x(t)\}_{t=1}^T$, along the direction vector X^{θ_k} , for all k (the whole set of direction vectors), giving $\{P^{\theta_k}(t)\}_{k=1}^K$ as the set of projections.
- III. Find the time instants $\{t_i^{\theta_k}\}_{k=1}^K$ corresponding to the maxima of the set of projected signals $\{P^{\theta_k}(t)\}_{k=1}^K$.
- IV. Interpolate $[t_i^{\theta_k}, s(t_i^{\theta_k})]$, for all values of k , to obtain multivariate envelope curves $\{e^{\theta_k}(t)\}_{k=1}^K$.
- V. For a set of K direction vectors, calculate the mean $\mu(t)$ of the envelope curves as

$$\mu(t) = \frac{1}{K} \sum_{k=1}^K e^{\theta_k}(t).$$

- VI. Extract the “detail” $d(t)$ using $d(t) = X(t) - \mu(t)$. If the “detail” $d(t)$ fulfills the stoppage criterion for a multivariate IMF, apply the above procedure to $X(t) - \mu(t)$ otherwise apply it to $d(t)$.

Basically, MEMD most significantly solve uniqueness problem associated with single channel EMD when processing multichannel data such as EEG signal.

III. COMMON SPATIAL PATTERNS

Firstly, H. Ramoser used common spatial patterns (CSP) technique to classify multi-channel EEG during imagined hand movements [12]. The main idea is to use a linear transform to project the multi-channel EEG data into low dimensional spatial subspace with a projection matrix of which each row consist of weights for channel. This transformation can maximize the variance of two class signal. CSP method is based on the simultaneous diagonalization of the covariance matrices of both classes.

Details of the algorithm are described as follows with the example of classifying EEG during hand and foot movements. Here, X_H and X_F denote EEG matrices during single trail hand and foot imagination with dimensions $N \times T$, where N is the number of channels and T is the number of samples per channel. The normalized spatial covariance of the EEG can be calculated from:

$$R_H = \frac{X_H X_H^T}{\text{trace}(X_H X_H^T)} \quad R_F = \frac{X_F X_F^T}{\text{trace}(X_F X_F^T)} \quad (1)$$

X^T is the transpose of X and $\text{trace}(X)$ computes the sum of the diagonal elements of X . The averaged normalized covariance $\overline{R}_H$ and $\overline{R}_F$ are calculated by averaging over all the trails of each group. The composite spatial covariance can be factorized as

$$R = \overline{R}_H + \overline{R}_F = U_0 \Sigma U_0^T \quad (2)$$

where U_0 the matrix of eigenvectors and Σ is the diagonal matrix of eigenvalues. The whitening transformation matrix

$$P = \Sigma^{-1/2} U_0^T \quad (3)$$

Transforms the average covariance matrices as

$$S_H = P \overline{R}_H P^T \quad S_F = P \overline{R}_F P^T \quad (4)$$

S_H and S_F share common eigenvectors and the sum of corresponding eigenvalues for the two matrices will always be the identity matrix,

$$S_H = U \Sigma_H U^T \quad S_F = U \Sigma_F U^T \quad \Sigma_H + \Sigma_F = I \quad (5)$$

The eigenvectors with the largest eigenvalues for S_H have the smallest eigenvalues for S_F and vice versa. The transformation of whitened EEG onto the eigenvectors corresponding to the largest eigenvalues in Σ_H and Σ_F is optimal for separating variance in two signal matrices. The projection matrix W is denoted as

$$W = U^T P \quad (6)$$

With the projection matrix W , the original EEG can be transformed into uncorrelated components

$$Z = W X \quad (7)$$

Z can be seen as EEG source components including common and specific components of different tasks. The original EEG X can be reconstructed by

$$X = W^{-1} Z \quad (8)$$

where W^{-1} is the inverse matrix of W and the columns of W^{-1} are spatial patterns which can be considered as EEG source distribution vectors. The first and last columns of W^{-1} are the most important spatial patterns that explain the largest variance of one task and the smallest variance of the other.

IV. DATA AND METHOD

A. Data Acquisition

The EEG data were collected from the BCI Competition III [13] dataset IVa. This data set was recorded from five healthy subjects (labeled aa, al, av, aw, ay). Each subject is performed right hand (class 1) and right foot (class 2) motor imaginary task. The subjects sat in a comfortable chair with arms resting on armrests. The data is recoded from 118 electrodes

according to the international 10/20 system with sampling frequency 100 Hz using BrainAmp MR plus amplifiers and an Ag/AgCl electrode cap. There were 280 trials for each subject, where 140 trials for each task. In each trial, visual cues are displayed and subjects imagined one of the two possible tasks for 3.5 seconds.

B. Proposed Algorithm

- 1) Channel selection: It is known that, center parts of the brain are most active during motor imageries [14]. In this experiment, we selected 13 channels “FC3,” “FC4,” “Cz,” “C1,” “C2,” “C3,” “C4,” “C5,” “C6,” “T7,” “T8,” “CP3,” and “CP4” according to the 10–20 system. We can call channel for short 34, 38, 54, 53, 55, 52, 56, 51, 57, 50, 58, 70, and 74.
- 2) Basic filtering: Filtering data can help remove high-frequency artifacts and low frequency floats. In this experiment, we applied a fifth-order Butterworth band-pass filter at first to each channel, which the pass-bands for each subject were 7–30 Hz. Therefore, we choose the alpha (7–12 Hz) and beta (13–30 Hz) bands, which are more effective during motor imagery task.
- 3) Filtering with MEMD: To decompose multi-channel EEG signal into a set of IMFs, we used MEMD. Now, we have to select IMFs, which will increase, our accuracy of classification. It is observed that first IMF carries noise and component that have frequency higher than 30Hz which carries the maximum energy among all frequencies found in the spectrum and last two IMFs are residue. Therefore, we reject first IMF and last two IMFs after decomposing the original signal $x(t)$ using MEMD. Now each channel has the same number of IMF index $c_j(t)$. In our new method, we remove first IMF and last two IMFs, thus a new signal $\hat{x}(t)$ is reconstructed by

$$\hat{x}(t) = \sum_{j=2}^{\Omega-2} c_j(t)$$

where Ω is the total number of IMF, which is same in each channel and t , represents time. Now, reconstructed signal $\hat{x}(t)$ carry more useful information than the original signal.

- 4) Feature Extraction & Classification: There have many techniques for feature extraction, among them we select common spatial filter (CSP) for this work. Output of CSP is a projection matrix W , the original EEG X can be transformed into uncorrelated components Z .

$$Z = WX$$

We choose an optimal subset of features, to improve classification performance. We choose the first row of Z as a feature. For classification, we used support vector machines with radial basis function (rbf) kernels and rbf sigma (σ)=10, through 10-fold cross-validation.

V. RESULTS AND DISCUSSION

In existing method [6], EMD and MEMD are used to analyze motor imagery response and CSP is used as a classifier. In another method [6] laplacian derivative is applied before EMD, MEMD or CSP by referencing an electrode to the mean of its four nearest neighboring electrodes. But all those methods have subject's dependency. Our proposed method increased averaged classification accuracy along with five subject and reduced subject dependency and relatively

stable than method at Long Zhang et al. Our processing framework serves to design more reliable MI-BCIs, which is of great interest for speeding up communication in real-life applications.

Table I shows that classification accuracy obtained by using proposed method. From TABLE I, it is clear that proposed algorithm achieves a better classification performance both individually and overall to subjects. The aim of motor imagery classification experiments is to identify IMFs which are bearing significant relevant data and that is achievable using MEMD for time-frequency analysis. Significant improvements for two class motor imagery dataset indicate that, we successfully extract the alpha and beta rhythms of interest. Cross validation is a model evaluation technique to estimating how well the model learned from some training data and what is going to perform on unseen data. In our experiments we used k-fold cross validation, which works by dividing the training set into k parts of equal size, then using $(k-1)$ parts as a training set and check the classification rate on the one remaining part. The performance of proposed method is evaluated with SVM classifier through a 10-fold cross validation method.

TABLE II
PERFORMANCE COMPARISONS USING PROPOSED METHOD

Method	Classification accuracy rate (%)					
	aa	al	av	aw	ay	Avg
Proposed method	82.0	92.2	69.4	83.8	89	83.3
CSP1 [6]	71.3	88.4	48.6	89.9	79.9	75.6
CSP2 [6]	65.3	90.2	63.7	80.3	87.3	77.4
EMD-CSP [6]	68.4	89.6	64.1	82.5	86.9	78.3
MEMD1-CSP [6]	68.8	90.0	68.8	76.3	87.5	78.3
MEMD2-CSP [6]	60.3	82.9	55.3	60.7	74.0	66.4
CSP1 with laplacian derivative [6]	70.7	78.3	48.7	88.5	72.6	71.8
CSP2 with laplacian derivative [6]	63.7	87.7	64.0	79.1	86.3	76.2
EMD-CSP with laplacian derivative [6]	62.8	89.5	64.4	79.6	85.9	76.4
MEMD1-CSP with laplacian derivative [6]	63.8	95.0	63.8	71.2	88.1	76.4
MEMD2-CSP with laplacian derivative [6]	55.3	79.7	54.7	70.0	75.6	67.1

VI. CONCLUSIONS

In this experiment, we focus on realization of a brain computer interface for the paralyzed people to control assistive environmental devices. For this purpose, motor imagery tasks are classified using different signal processing methods. For filtering frequencies which are not related with motor imagery by basic band-pass filter and multivariate empirical mode decomposition (MEMD) to enhance the EEG signal. Unlike the standard single channel EMD algorithm, MEMD allowed a more stable estimation of the time-varying frequency response from multichannel EEG and preserve the cross channel interdependence. CSP is used as discriminating tool that helps to differentiate different mental status. In this experiment, we use support vector machine (SVM) as a classifier with a non-linear rbf kernels. This study developed a promising method based on MEMD-CSP aid with SVM that shows better performance in MI based classification.

ACKNOWLEDGMENT

This work is supported by ICT fellowship-2015-16, information and communication technology division, govt. of the people's republic of Bangladesh.

REFERENCES

- [1] M. Grosse-Wentrup, C. Liefhold, K. Gramann, and M. Buss, *Beam forming in noninvasive brain-computer interfaces*, IEEE Trans. Biomed. Eng., vol. 56, no. 4, pp. 1209–1219, Apr. 2009.
- [2] G. Pfurtscheller, C. Brunner, A. Schlogl, and F. Lopes da Silva, “*Mu rhythm (de) synchronization and EEG single-trial classification of different motor imagery tasks*,” Neuroimage, vol. 31, no. 1, pp. 153–159, 2006.
- [3] Le Quoc Thang, Chivalai Temiyasathit, Increase performance of four-class classification for Motor-Imagery based Brain-Computer Interface.
- [4] BAI Xiaoping, WANG Xiangzhou, ZHENG Shuhua, YU Mingxin, *The Offline Feature Extraction of Four-class Motor Imagery EEG Based on ICA and Wavelet-CSP*, Proceedings of the 33rd Chinese Control Conference July 28–30, 2014, Nanjing, China.
- [5] M Agustina Garcés Correa and Eric Laci Leber. Noise removal from eeg signals in polisomnographic records applying adaptive filters in cascade. 2011.
- [6] Long Zhang, Cheng Zhang, Hiroshi Higashi, Jianting Cao, Toshihisa Tanaka, *Common Spatial Pattern Using Multivariate EMD for EEG classification*, APSIPA ASC 2011 Xi'an.
- [7] J. Müller-Gerking, G. Pfurtscheller, and H. Flyvbjerg, *Designing optimal spatial filters for single-trial EEG classification in a movement task*, Clinical Neurophysiology, vol. 110, pp. 787–798, 1999.
- [8] H. Ramoser, J. Müller-Gerking, and G. Pfurtscheller, *Optimal spatial filtering of single trial EEG during imagined hand movement*, IEEE Trans. Rehabil. Eng., vol. 8, no. 4, pp. 441–446, 2000.
- [9] Cheolsoo Park, David Looney, Naveed ur Rehman, Alireza Ahrabian, and Danilo P. Mandic, *Classification of Motor Imagery BCI Using Multivariate Empirical Mode Decomposition*, IEEE Trans. on Neural Systems and Rehabilitation Eng., vol. 21, no. 1, January 2013.
- [10] N. Rehman and D. P. Mandic, *Multivariate empirical mode decomposition*, Proc. R. Soc. A, vol. 466, no. 2117, pp. 1291–1302, 2010.
- [11] H. Drucker, C. J. C. Burges, L. Kaufman, A. J. Smola, and V. Vapnik, *Support vector regression machines*, M. C. Mozer, M. I. Jordan, and T. Petsche, Eds. Cambridge, MA: MIT Press, 1997.
- [12] H. Ramoser, J. Müller-Gerking and G. Pfurtscheller, *Optimal spatial filtering of single trial EEG during imagined hand movements*, IEEE Trans. Rehab. Eng., vol. 8, no. 4, pp. 441–446, 2000.
- [13] BCI Competition III [Online]. Available: <http://www.bbc.de/competition/iii>
- [14] C. Neuper, G. R. Müller-Putz, R. Scherer, and G. Pfurtscheller, *Motor imagery and EEG-based control of spelling devices and neuroprostheses*, Progress in Brain Research, vol. 159, pp. 393–409, 2006.